

Publication-ready figures

LaTeX fonts, physical size, and accessibility

A reproducible `contrastcolors` demonstration

Abstract

This two-column article demonstrates a document-aware Matplotlib workflow. Figures are generated from dimensions and font sizes measured from this LaTeX document. One graph is included using PGF, so its text is typeset with the same LaTeX font configuration as the surrounding paragraphs. The remaining examples illustrate a vector PDF, rasterized scatter artists, and six simulated accessibility conditions.

1 What does font matching mean?

Scientific plots should be designed at their *final physical width*. The preamble probe measures `\columnwidth`, `\textwidth` and normal body font size. In these plots, axis labels use the body point size; tick and legend labels follow a consistent hierarchy.

A PDF produced independently by Matplotlib can have the correct font *size* without matching its glyphs. PGF instead instructs LaTeX to typeset text when the figure is included.

2 Grouped scatter

A categorical plot should not rely on hue alone. Figure 3 assigns each category a distinct color and marker. All observations in one category keep the same marker. Three groups use broadly spaced relative luminances so their grayscale variants remain visibly separated. Legends sit above the data rather than covering points.

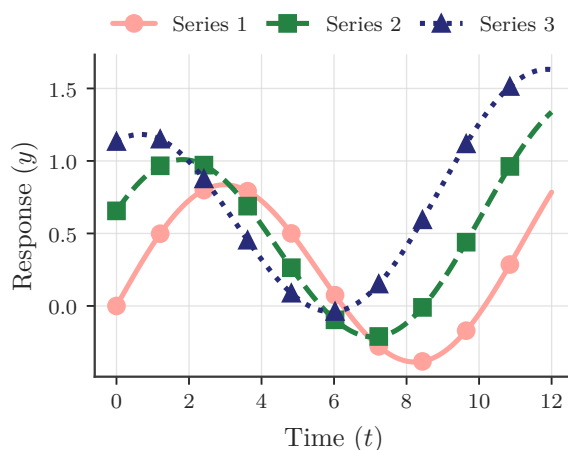


Figure 1: **Native PGF**. Text and math labels use the font configuration of this document. Markers and line patterns identify the plotted series in addition to their colors.

3 Hybrid vector and raster export

Large scatter clouds can make vector PDFs unnecessarily heavy. Figure 4 rasterizes dense points but retains the surrounding axes, labels and ticks as vectors.

4 Accessibility

A color-vision simulation is a diagnostic approximation, not a clinical prediction or universal certification. Figure 5 compares the same line plot in six views: original color, luminance-preserving grayscale, print stress, deuteranopia, protanopia, and tritanopia. Each view has a separate header and is displayed in a two-column grid to preserve

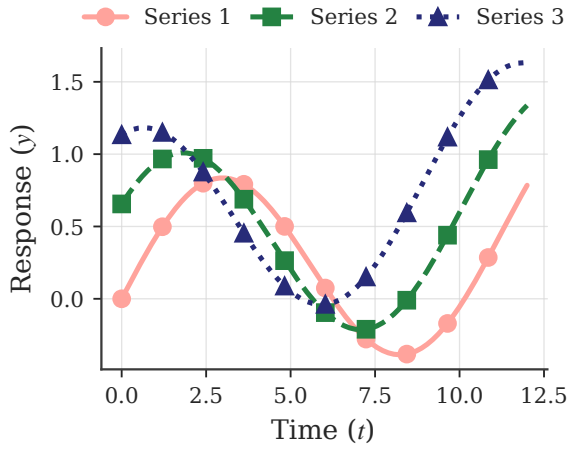


Figure 2: **Direct PDF export.** The same plot at the same physical column width. The nominal font size is correct, though the glyph outlines can differ from Figure 1.

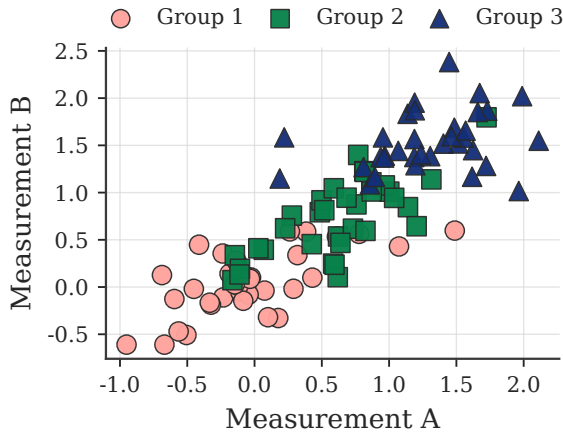


Figure 3: Categorical scatter with redundant color and marker encoding, exported at the manuscript's actual column width.

legibility at the final printed size.

5 Reproducibility

The accompanying script uses a fixed random seed, saves a JSON report with font lookup, physical widths, and audit results, and compiles a standalone proof confirming that the PDF figure's un-

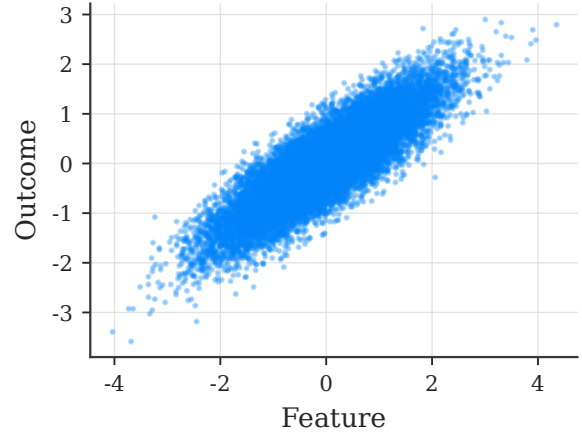


Figure 4: **Hybrid rendering.** Dense data points are rasterized, while simple graphic elements and typography remain vectorial.

scaled width matches the measured LaTeX column. Matching nominal point size, using PGF, and inspecting grayscale do not eliminate a final visual review of dense legends, labels and scientific clarity.

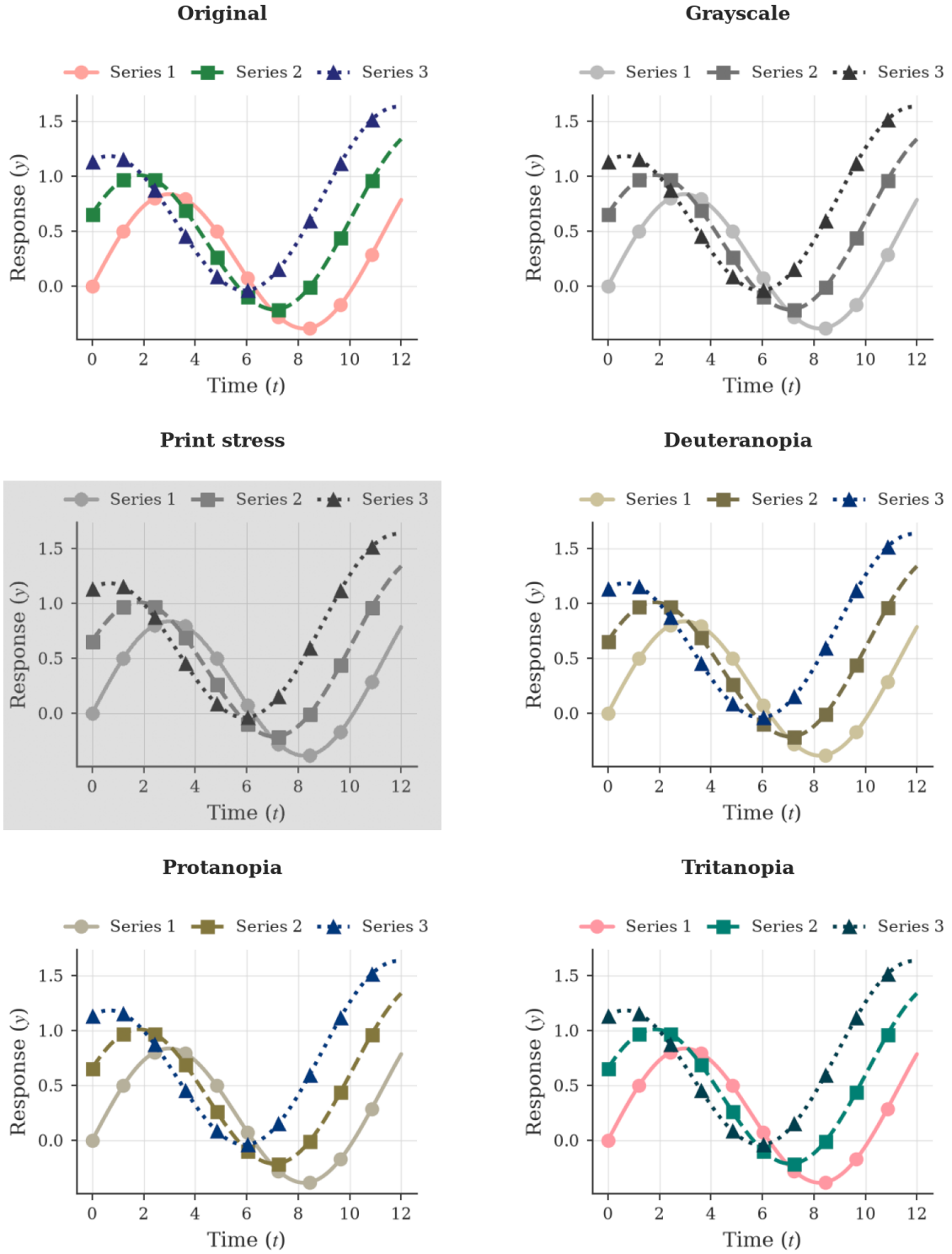


Figure 5: **Six-way comparison.** Original, black-and-white, degraded print, and three Machado-model color-vision-deficiency previews. Headers are outside the image areas. The panel is rasterized; the source PDF and PGF above retain appropriate vector content.

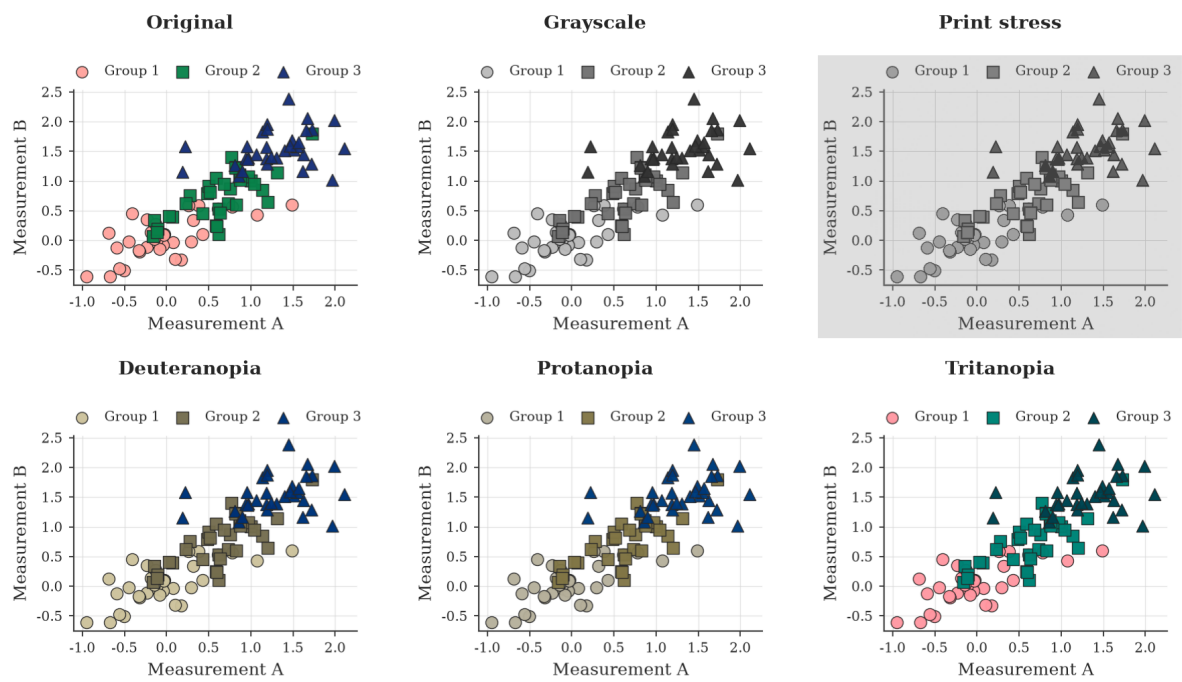


Figure 6: **Scatter accessibility panel.** The same categorical groups remain identified by their symbols in black and white and in simulated color-vision-deficiency conditions.